

# Marketing Campaign EDA Summary Report

## Overview

This project focused on performing Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand customer behavior, campaign effectiveness, and return on investment (ROI). The dataset was cleaned and analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn.

The objective of the analysis was to identify which marketing channels perform best and recommend where the company should allocate more marketing budget.

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## Data Cleaning Process

The dataset was first preprocessed by:

- Handling missing values
- Removing duplicate records
- Checking data types
- Creating new columns such as Total Spending
- Preparing the dataset for analysis and visualization

This ensured accurate and reliable analysis results.

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## Key Insights from Analysis

### 1. Customer Spending Behavior

Customers with higher income levels were found to spend more on products and marketing campaigns. Income showed a positive relationship with total spending.

### 2. Best Performing Marketing Channel

Among all purchase channels:

- Web purchases generated the highest number of transactions
- Store purchases performed moderately well
- Catalog purchases had comparatively lower engagement

This indicates that digital marketing channels are more effective in reaching customers.

### **3. Campaign Response Analysis**

The majority of customers did not respond to the campaign, but a smaller group of customers showed high engagement and purchasing activity. Targeting these high-response customers can improve future campaign performance.

### **4. Customer Engagement Trends**

Customers who previously accepted campaigns were more likely to respond positively again. This suggests that customer loyalty and repeated engagement play an important role in marketing success.

### **5. Correlation Analysis**

The correlation heatmap showed strong relationships between income, spending, and purchase behavior. Customers with higher spending patterns were also more active across multiple purchase channels.

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## **Recommendations**

Based on the analysis, the following recommendations are suggested:

- Increase budget allocation for web and digital marketing campaigns
  - Focus on high-income customer segments
  - Improve personalized marketing strategies for repeat customers
  - Use customer behavior data for targeted advertisements
  - Reduce spending on less effective marketing channels
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## **Conclusion**

The Exploratory Data Analysis helped identify important customer and marketing trends. The results showed that digital marketing channels provide better ROI and customer engagement compared to traditional channels. These insights can help businesses make data-driven marketing decisions and improve campaign effectiveness in future marketing strategies.